



Program Specification of Master in Dental Sciences (MDS)

الاسم التوقيع	ممثل البرنامج د/ خالد أحمد الجوفي	اسم المراجع أ.د. نعمان النجار	رئيس القسم ومنسق البرنامج د/ محمد نادر قلعة جي	عميد الكلية د/ خالد أحمد الجوفي

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إدارة ضمان الجودة والاعتماد - رئاسة الجامعة - ت : 373240 فاكس : 4144 / 4155

تسعى إدارة ضمان الجودة والاعتماد إلى التمسك بالمستمرات العلمية وبرامج الجامعة، ولذا تلتزم الجودة العلمية العالمية، وإعلاء قيم التعليم، وتطبيق معايير الاعتماد.

Introduction:

Faculty of Dentistry at University of Science and Technology offers a Master program in Dental Sciences (MDS). It is designed to build on the undergraduate degree experience and continuous to participation with Faculty of Dentistry, Egyptian Tanta University that ranged from 2008 until 2012 in which two batches are graduated (about 30 student in deferent dental specialties). It further enhance student critical thinking and research skills. The scope of the degree to improve health care outcomes and solve dental community problems. Therefore, this program is determined into five tracks are: Oral Medicine & Periodontology, Fixed & Removable Prosthodontics, Endodontics & Operative Dentistry, Oral Surgery & Implantology, and Orthodontics.

Program Identification and General Information:

1	Program title: Master in Dental Sciences (MDS).
2	Program Type: Single Program () Joint Program () Multidisciplinary Program (☒)
3	Number of years needed for completion of the program: 2 - 3 Years.
4	Total credit hours needed for completion of the program: 39 C. H.
5	Award granted on completion of the program: Master in Dental Sciences –Track name.
6	Name of the university which provide the Award: University of Science and Technology
7	Name of the faculty which own the program and provide the Award: Faculty of Dentistry
8	Department which own the program: 1- Department of Biological & Preventive Sciences. 2- Department of Prosthodontics and Conservative Dentistry.
9	Name of the faculty which participates in teaching some of the program courses: Faculty of Medicine
10	Date of program development: 2014-2015
11	Program Coordinator: Dr. Khaled Ahmed AL-Jawfi
12	External Reviewer\ External Reviewers:
13	Date of Program Approval from the university and the MoHE: - Department Council: - Faculty Council: - Deans Council: - University Council:

جامعة العلوم والتكنولوجيا
إدارة ضمان الجودة والإعتماد
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University Vision, Mission and Goals:

University Vision:

To become one of the leading and distinguished universities regionally and internationally.

University Mission:

The University of Science and Technology strives to provide an excellent education service, enhancing student knowledge and skills that meet the requirements of the labor market, contributing to community development by providing training, advisory, and research services and the continuing education programs through competent administrative and academic staff, stimulating learning environment, and cutting-edge programs. The University shall expand local, regional, and international partnerships according to the comprehensive quality systems, transparency, and the professional ethics.

University Goals:

1. Enhance student knowledge and skills in various academic disciplines.
2. Contribute to support efforts of practical and scientific research in all different areas.
3. Link the learning outcomes with the requirements of the labor market to ensure the graduates are employable
4. Provide the necessary infrastructure to support the learning process and students' services and activities.
5. Provide opportunities for education in separate locations for male and female students.
6. Expand the partnerships and develop the relationships with other universities and institutions of scientific research domestically, regionally, and internationally.

Faculty Vision:

To achieve the leadership in dentistry through producing competent dentists at the regional and global level.

Faculty Mission:

The Dental College strives to be one of the premier colleges at the local, regional and international level, through raising the performance level in the academic and administrative aspects, providing conducive environment for learning and training and conducting practical research, as well as upscale therapeutic services to the community. It also seeks to produce competent dentists capable to apply their skills and experiences that they have learned with efficiency.

Faculty Goals:

1. Provide high level of education and training to ensure rehabilitation of dentists with high standards of efficiency.
2. Develop the curricula and course structures to ensure the student's involvement in the self-learning process and development of continuing education to meet the standards of quality and accreditation.
3. Provide oral and dental health service to the people through dental clinics in the college.
4. Develop skills of faculty members to keep pace with the developments in the methods of teaching and learning process.
5. Activate the partnership and development of relations with the other dental colleges/peers and institutions for scientific research locally, regionally and internationally.
6. Connect with the graduates of the college and link them with the college to develop their research and professional expertise.

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التوقيع	د/ خالد أحمد الجوفي	د/ نعيان النجار	د/ محمد نادر قلعة جي	د/ خالد أحمد الجوفي

Program Vision, Mission and Goals

• Program Vision

It is to be regionally and internationally distinguished as pioneer in higher dental education , health care , and collaborative community oriented research.

• Program Mission

The master program in dental sciences intents to develop academic and research student skills, and to improve dental community services.

• Program Goals

The master program in dental sciences aims to:

- 1- Acquire students an advanced knowledge of specialist fields in dental sciences.
- 2- Enhance students' critical, analytical, practical and communication skills relevant to an advanced specialist field.
- 3- Develop research skills in terms of : planning, conducting, evaluating and reporting the results of investigations.
- 4- Link the labor market requirements with oral and dental services to solve the dental community problems.

Intended Learning Outcomes (ILOs):

They are prepared according to several protocol procedures such as survey of intended learning outcomes in five similar programs, survey of the requirements in labor market, and workshop to discuss the suggested outcomes.

Intended Learning Outcomes (ILOs) classify into:

A. Knowledge and Understanding

Program Outcomes The student will be able to:	Teaching & Learning Methods	Assessment Methods
A1. Demonstrate understanding of advanced and recent knowledge in dental specialist fields.	-Lectures. -Seminars. -Self learning. -Dialogue and discussion. -Course work assignment.	-Written examination (MCQs, essays, etc...). -Oral examination. -Notes and observation.
A2. Show knowledge and understanding of the bio-medical, para-clinical, clinical and behavioral disciplines that are relevant to advanced dentistry.		
A3. Recognize the fundamental principles , ethics, and research methodology in dental specialist fields.		
A4. Identify the modern scientific developments related to professional practice in dental specialist fields.		

B. Intellectual Skills

Program Outcomes The student will be able to:	Teaching & Learning Methods	Assessment Methods
B1. Analyze critically knowledge of basic sciences and scientific literature relevant to dental specialist fields.	-Lectures. -Seminars. -Self learning.	-Written examination (MCQs, essays, etc...). -Oral examination.
B2. Evaluate the effect of treatment on the practical		

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and/or clinical competence necessary to carry out appropriate treatment in dental specialist fields.	- Dialogue and discussion.	- Notes and observation.
B3. Conduct research skills as ability to appraise scientific papers and write research report adhering to the appropriate ethical codes.	- Problem solving.	- Evaluation assignment.
B4. Formulate research problems using the scientific methods and modern developments in dental specialist fields.	- Coursework assignment.	
	- Case study analysis.	
	- Brain storming.	

C. Professional and Practical Skills

Program Outcomes The student will be able to:	Teaching & Learning Methods	Assessment Methods
C1. Use adequate professional basic skills and recent advances in performing various procedure required in dental specialist fields.	- Seminars.	- Evaluation of reports.
C2. Employ the appropriated clinical examinations using contemporary concepts, techniques and materials used in the diagnosis and treatment planning of patients requiring specialist field.	- Clinical problem based-solving scenarios.	- Clinical and /or practical examinations.
C3. Apply standard visual, technical and manual examinations with evaluation and development the techniques used in the field of specialty.	- Routine laboratory and clinical work.	- Evaluation of case study and diagnosis.
C4. Explore at depth a specific research area and write a dissertation on the subject applying high moral and ethical standards while carrying out human or animal research.	- Case study.	- Evaluation of practical and/or clinical steps.
	- Assignments.	- Notes and observation.
	- Tutorials.	

D. General Skills

Program Outcomes The student will be able to:	Teaching & Learning Methods	Assessment Methods
D1. Manage time and communicate effectively using a variety of formats and a range of information sources.	- Problem solving.	- Oral examinations.
D2. Work effectively alone (e.g. on assignments or during the project) and as part of a team (e.g. in group work, during group discussions and workshops).	- Seminars.	- Evaluation of reports.
D3. Develop the skills required for continuing self-managed professional development and learning.	- Cooperative learning.	- Observations.
D4. Respect the ethical issues related to the scientific researches.	- Self learning.	- Evaluation of presentation.
	- Dialogue and discussion.	
	- Course work assignment.	



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Mapping PILO's with the benchmarks

	PILOs	Standards and Benchmarks*
Knowledge & Understanding	A1	(London & Indian Universities) & (Labor Market, Students, Graduates)
	A2	(London & Bergen Universities) , ADA, and (Labor Market, Students, Graduates)
	A3	(Malaya & Cairo Universities) & ADA
	A4	(London & Cairo Universities) , ADA, and (Labor Market, Students, Graduates)
Intellectual (Cognitive) Skills	B1	(Malaya & Bergen Universities) , ADA, and (Students , Graduates)
	B2	(Bergen & Indian Universities) & (Labor Market, Students, Graduates)
	B3	(Malaya & Indian Universities) , ADA, and (Students, Graduates)
	B4	(London & Cairo Universities) , ADA, and (Students, Graduates)
Professional and Practical Skills	C1	(Cairo & Indian Universities) & (Labor Market, Students, Graduates)
	C2	(Malaya & London Universities) , ADA, and (Labor Market, Students, Graduates)
	C3	(Bergen & Cairo Universities) , ADA, and (Labor Market, Students, Graduates)
	C4	(London & Indian Universities) , ADA, and (Labor Market, Students, Graduates)
General Skills	D1	(Malaya & Indian Universities) , ADA, and (Labor Market, Students, Graduates)
	D2	(London & Cairo Universities) & (Labor Market, Students, Graduates)
	D3	(Malaya & Indian Universities) & , ADA, and (Labor Market, Students, Graduates)
	D4	(Malaya & Cairo Universities) , ADA, and (Students, Graduates)

Standards and Benchmarks are depended on:

Accreditation standards for advanced specialty education programs in dental sciences , Commission on Dental Accreditation, American Dental Association (ADA).

Intended Learning Outcomes are prepared from certain universities, while tracks, courses and credit hours are prepared from other universities.

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Program Standards:

This section introduces academic standards, benchmarks and other references which include labor market needs and academic staff. These references are as follows:

■ Academic Standards:

Program developed according to standards of Commission on Dental Accreditation that followed to American Dental Association. (Annex 1) ADA Criteria for Dental Education Programs.

■ Benchmarks:

The development of Master Program in Dental Sciences is depended on many famous universities. See: (Annex 2) Similar programs names and information.

(Annex 3) Survey of Intended Learning Outcomes for Similar Accredited Programs at International Universities.

(Annex 4) Survey of credit hours for programs similar to the current program.

(Annex 5) Survey of courses for Similar Accredited Programs at International Universities).

■ Academic Staff:

A group of Teaching Staff Members in different knowledge fields of scientific contents participate program specification document development. (Annex 6) Academic Staff Names (curriculum development committee) and other staff involved in PSD development process.

Intended Learning Outcomes Mapping:

For achievement the balance and catholicity in the courses from their coverage to different scientific themes of the program, the major and minor scientific fields and their relative weights was determined (Annex 7), also doubtlessness from linking process between these themes and learning outcomes that covered (Annex 8), then linking of the themes and study courses (Annex 9), and finally the learning outcomes of study courses were mapped according to (Annex 10).

Annex (7): Main Themes/Sub-Themes with Relative weight for Program.

Annex (8): PILOs Distribution to General Themes for Program.

Annex (9): Course Titles and p- ILOs Distribution to General Themes for Program.

Annex (10): Matrix of mapping program P- ILO's with courses.

Program Structure / Plan

Duration of the program: 2 – 3 Years.

Total credit hours: Thirty Nine Credit Hours.

Program delivery mode: Credit Hours System.

Study Plan Framework

Requirements	Credit Hours
1. Compulsory Basic Courses	12
2. Compulsory Specialty Courses	9
3. Elective Specialty Courses	9
4. Master Thesis in Specialty	9
Total of Credit Hours	39

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1. Compulsory Basic Courses (12 credit hours for all tracks):

Course Title	Code NO.	No. of credit hours			Pre-Requisites
		Th.	Pr.	Total	
1 Biostatistics	MDS001	3	0	3	
2 Research Methodology	MDS002	3	0	3	
3 Applied Head & Neck Anatomy	MDS003	2	1**	3	
4 Human Physiology	MDS004	2	1**	3	
Total Credit Hours		10	2	12	

2. Compulsory Specialty Courses (9 credit hours for all tracks):

Course Title	Code NO.	No. of credit hours			Pre-Requisites
		Th.	Pr.	Total	
1 Advanced Oral Histology	MDS005	2	1**	3	
2 Advanced Oral & Maxillofacial Pathology	MDS006	2	1**	3	
3 Advanced Oral Microbiology & Immunology	MDS007	2	1**	3	
Total Credit Hours		6	3	9	

3.1 Elective Specialty Courses (9 credit hours for Oral Medicine & Periodontology Track):

Course Title	Code NO.	No. of credit hours			Pre-Requisites
		Th.	Pr.	Total	
1 Advanced Oral Medicine	MDS008	2	1***	3	
2 Advanced Periodontology	MDS009	2	1***	3	
3 Advanced Oral & Maxillofacial Radiology	MDS010	2	1**	3	
4 Advanced Oral Diagnosis	MDS011	2	1**	3	
5 Applied Dental Biomaterials	MDS012	2	1**	3	
Total Credit Hours		10	5	15	

3.2 Elective Specialty Courses (9 credit hours for Fixed & Removable Prosthodontics Track):

Course Title	Code NO.	No. of credit hours			Pre-Requisites
		Th.	Pr.	Total	
1 Advanced Fixed Prosthodontics	MDS013	2	1***	3	
2 Advanced Removable Prosthodontics	MDS014	2	1***	3	
3 Advanced Oral & Maxillofacial Radiology	MDS010	2	1**	3	
4 Advanced Oral Diagnosis	MDS011	2	1**	3	
5 Applied Dental Biomaterials	MDS012	2	1**	3	
Total Credit Hours		10	5	15	

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3.3 Elective Specialty Courses (9 credit hours for Endodontics & Operative Dentistry Track):

Course Title	Code NO.	No. of credit hours			Pre-Requisites
		Th.	Pr.	Total	
1 Advanced Endodontics	MDS015	2	1***	3	
2 Advanced Operative Dentistry	MDS016	2	1***	3	
3 Advanced Oral & Maxillofacial Radiology	MDS010	2	1**	3	
4 Advanced Oral Diagnosis	MDS011	2	1**	3	
5 Applied Dental Biomaterials	MDS012	2	1**	3	
Total Credit Hours		10	5	15	

3.4 Elective Specialty Courses (9 credit hours for Orthodontics Track):

Course Title	Code NO.	No. of credit hours			Pre-Requisites
		Th.	Pr.	Total	
1 Advanced Removable Orthodontics	MDS017	2	1***	3	
2 Advanced Fixed Orthodontics	MDS018	2	1***	3	
3 Advanced Oral & Maxillofacial Radiology	MDS010	2	1**	3	
4 Advanced Oral Diagnosis	MDS011	2	1**	3	
5 Applied Dental Biomaterials	MDS012	2	1**	3	
Total Credit Hours		10	5	15	

3.5 Elective Specialty Courses (9 credit hours for Oral & Maxillofacial Surgery Track):

Course Title	Code NO.	No. of credit hours			Pre-Requisites
		Th.	Pr.	Total	
1 Advanced Oral Surgery	MDS019	2	1***	3	
2 Advanced Dental Implantology	MDS020	2	1***	3	
3 Advanced Oral & Maxillofacial Radiology	MDS010	2	1**	3	
4 Advanced Oral Diagnosis	MDS011	2	1**	3	
5 Applied Dental Biomaterials	MDS012	2	1**	3	
Total Credit Hours		10	5	15	

4. Master Thesis in Specialty (9 credit hours for each track):

Track	Course Title	Code No.	Credit Hours
All Tracks	Master Thesis in specialty	MDS021	9

** 1 credit hour = 2 contact hours.

*** 1 credit hour = 6 contact hours.

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Program Study Plan

1st Semester : (12 credit hours) : الفصل الأول :

Course Title	Code NO.	No. of credit hours			Pre-Requisites
		Th.	Pr. (Contact Hours)	Total	
1 Advanced Oral Histology	MDS005	2	2	3	
2 Applied Head & Neck Anatomy	MDS003	2	2	3	
3 Human Physiology	MDS004	2	2	3	
4 Biostatistics	MDS001	3	0	3	
Total Credit Hours		9	6	12	

2nd Semester : (9 credit hours) : الفصل الثاني :

Course Title	Code NO.	No. of credit hours			Pre-Requisites
		Th.	Pr. (Contact Hours)	Total	
1 Advanced Oral & Maxillofacial Pathology	MDS006	2	2	3	
2 Advanced Oral Microbiology & Immunology	MDS007	2	2	3	
3 Research Methodology	MDS002	3	0	3	
Total Credit Hours		7	4	9	

3rd Semester : (9 credit hours) : الفصل الثالث :

Course Title	Code NO.	No. of credit hours			Pre-Requisites
		Th.	Pr. (Contact Hours)	Total	
1 Elective (1)	-----	2	6	3	
2 Elective (2)	-----	2	6	3	
3 Elective (3)	-----	2	2	3	
Total Credit Hours		6	14	9	

4th Semester : (9 credit hours) : الفصل الرابع :

Track	Course Title	Code No.	Credit Hours
All Tracks	Master Thesis in specialty	MDS021	9



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Admission Requirements for the Program:

The candidate has to make an application to the Deanship of Postgraduate Studies & Scientific Research with the following documents along with the prescribed fee:

- 1- General standards according to Postgraduate Studies System.
- 2- BDS pass / degree certificate with a grade of a Good at least and academic record issued by University or any university is a recognized by the Ministry of Higher Education and Scientific Research with regarding the recognition of documents by it.
- 3- Completion of paid rotatory internship certificate from a recognized college.
- 4- Students for MDS shall be admitted based on the competitive procedures held by the scientific departments at Dental College.
- 5- Candidates should obtain the Eligibility Certificate before the last date for admission as notified by the Deanship of Postgraduate Studies & Scientific Research.
- 6- A candidate who has been admitted to MDS should register his / her name in the Deanship of Postgraduate Studies & Scientific Research within a week of admission after paying the registration fee.

Progress and completion conditions/requirements in the program

- 1- The Program shall be of 2-3 years duration.
- 2- Every candidate shall have not less than 80 percent of attendance in each semester of course. However, candidates should not be absent continuously as the course is a full time one.
- 3- Every candidate shall maintain a work diary and record of his/ her participation in the training program conducted by the department such as , seminars, practical and/or clinical requirements, etc.
- 4- Records and marks obtained in tests will be maintained by the Deanship of Postgraduate Studies & Scientific Research and will be made available to the University when called for.
- 5- Every candidate pursuing MDS degree course is required to carry out work on a selected research project under the guidance of a recognized post graduate teacher. The results of such a work shall be submitted in the form of a dissertation.
- 6- Discussion & acceptance of dissertation shall be a condition for the candidate to finish the requirements of MDS degree.

Teaching and Learning resources for the program:

- 1- Library Resources: Tradition Books – Local & International Journals.
- 2- Electronic Resources: Internet – Electronic Books & Journals (CD-DVD).
- 3- Training and practice in the dental laboratories and clinics.
- 4- Biomedical research center at university hospital for research procedures.



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Facilities

- 1- Program delivery mode is a credit hours system.
- 2- All policies and procedures that relevant to MDS program are present and evident.
- 3- Teaching & learning resources , and student services are available.
- 4- Activate the partnership and development of relations with the other dental colleges and institutions of scientific research locally, regionally and internationally in order to develop the scientific research and professional expertise.
- 5- Infrastructure to support and service the program as follows:

Basic Structure	Number
Teachers	16
Technicians	12
Dental Clinics	7
Dental Laboratories	14
Teaching Halls	10

Program Assessment:

Type of the Sample who Assess the program	Instruments used	Sample
1- Final Year students	Questionnaire	Not yet
2- Graduates from the program	Questionnaire	Not yet
3- Employer	Questionnaire	20%
4- External evaluators	Multi disciplines	Not yet
5- Others	Multi disciplines	Not yet

Program Quality Standards:

- 1- Internal & External periodic evaluation.
- 2- Providing executed procedures to ensure program quality.
- 3- Preparing program self-study evaluation.
- 4- Execute assessing process to ensure program outcomes quality.
- 5- Evaluation of program from external committee.
- 6- Preparing executed corrected plan.
- 7- Build strong relationships between program and accredited organizations.

Internal and external training to satisfy program standards:

- 1- Workshops and training courses for academic staff (Teaching methods, Medical Education, Diploma of Professional Development Program for Teaching Staff in Higher Education,...etc).
- 2- Apply the modern training methods in dental laboratories and clinics.
- 3- Commutative training through activate the partnership and development of relations with the other dental colleges and institutions of scientific research locally, regionally and internationally.
- 4- Continued field visits

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